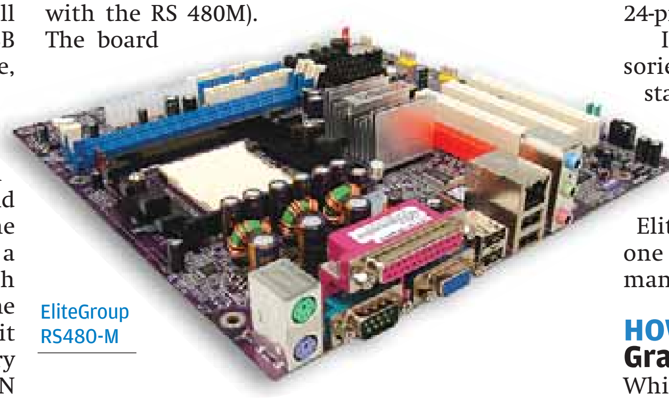
dard feature, with eight-channel getting there really fast!

In terms of connectivity options, the MSI RS480M2-IL was slightly ahead of the rest. Like all the others, it featured four USB ports with support for four more, but what it also had, unlike the others, was an onboard FireWire port. The other boards had support for FireWire in terms of pins on the board, but did not have a physical port on the back panel. The board featured a 10/100 Mbps Ethernet port, which was disappointing, as a lot of the other boards provided Gigabit Ethernet. Gigabit Ethernet is very helpful if you are working in a LAN environment, as data transfers would be, theoretically at least, much faster.

The falling prices of storage have not bypassed SATA drives, and these, too, are witnessing frequent price cuts along with capacity increases. We were amazed to see that every board, apart from the Krypton M7VIG400, provided support for at least two SATA connectors—and half the boards had four SATA connectors!

The features list doesn't end here. Half our boards also supported RAID in 0, 1, and 0+1 configurations. The MSI k8mmv did not support RAID, and hence lost out on the features front.

In terms of pure features, the most feature-rich was the MSI RS480M2-IL (not to be confused with the RS 480M). The board is based on PCIe, and features support for 4 GB of RAM, has three PCI slots, 5.1 channel integrated audio, four USB ports and even a FireWire port!


EliteGroup RS480-M

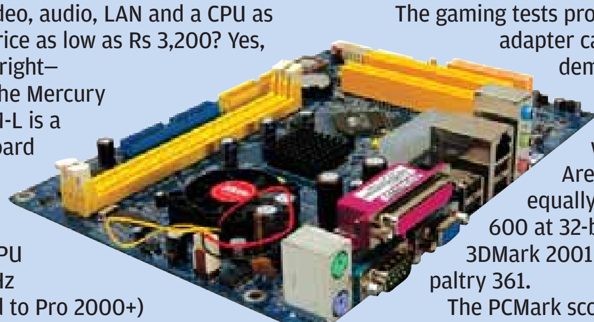
It features SATA with RAID (0, 1, 0+1) as well as 10/100 Mbps LAN. The notable thing here is that this board (as well as quite a few of our other boards) requires a 24-pin power supply instead of the regular 20-pin, so you need to be careful when purchasing the power supply.

There is nothing dangerous about plugging a 20-pin power supply into a 24-pin socket, except that you might not be able to give the board enough power, and might not be able to add other components later due to lack of power—so it's wise to purchase a 24-pin power supply at the outset.

In terms of bundled accessories, all boards came with the standard cables and connectors except the Gigabyte, which also bundled in a USB LED (for fun, we presume). The other exception was the Elite Group RS480M board. This one came with nothing—no manuals, no cables.

The Mercury PVCLE266M-L

Would you believe us if we told you there's a motherboard available with onboard video, audio, LAN and a CPU as well, at a price as low as Rs 3,200? Yes, you read it right—Rs 3,200! The Mercury PVCLE266M-L is a Mini ATX board that comes with the VIA C3 Samuel 2 CPU (CX 800 MHz overclocked to Pro 2000+) onboard. The CPU operates at an FSB of 133 MHz. The board is based on the VIA CLE266 chipset. There is support for two DDR memory modules running at up to 266 MHz. As much as 2 GB of memory is supported.



There are two IDE connectors that can support a maximum of four IDE devices with speeds up to ATA133. Four USB2.0 ports are provided. Connectivity is via a 10/100 LAN onboard.

The onboard VGA is a S3 display adapter with primary 3D functions. The audio device is an AC'97 supporting 6 channels.

Expansion slots are provided in the form of 2 PCI and one CNR slot.

The gaming tests proved that the display adapter cannot aptly handle demanding 3D games. Serious Sam logged a low 5.6, while Quake 3 Arena logged an equally low 17.9 at 800 x 600 at 32-bit colour. The 3DMark 2001 SE score was a paltry 361.

The PCMark score was 451, which is 15 per cent less than that of most other boards we tested. In the SiSoft Sandra CPU benchmark, the CPU was quite low, and the memory benchmark was no different—5.5 in the ZDBench Business Winstone.

The above tests prove that while the motherboard is not well-suited for 3D gaming at all, it can still be used in offices where work is mostly limited to spreadsheets and word processors. It can also be used by cyber-cafés and computer classes, where its specifications would be more than sufficient to get the job done.

HOW THEY FARED

Graphics Performance

While testing these boards, we kept in mind that the target audience will not be looking at a gaming solution, so we ran less demanding games (Quake 3 and Serious Sam) at lower resolutions (640 x 480 and 800 x 600).

The boards with onboard graphics did a fairly commendable job, and gave more than playable frame rates in most cases (assuming anything above 30 fps is playable).

In Serious Sam, the lowest scores were logged by the Asus K8SMX-UAYZ. This SiS 760GX chipset-based motherboard could just manage a paltry 26 fps at a low resolution of 640 x 480 and similar 22.6 at 800 x 600.

Motherboards Without Onboard Graphics

The newer chipsets from nVidia do not feature support for onboard video. Even the earlier version of nForce3 lacked onboard video and came with an AGP slot. The motherboards based on the nForce4 chipset were the Gigabyte (the GA-K8NE) and the two WinFast boards, the NF4K8MC and NF4K8AB.

After a long sabbatical, ULi (earlier ALi) has come back with a brand new chipset for AMD boards (remember how, earlier, ALi and Aladdin were the chipsets AMD users swore by? Those guys are back!)—the ULi M1689. The board is AGP-based, and does not have onboard video, which makes us think that Ali is a little behind the times here (PCIe is here, guys!).

Moving on and coming to their graphics performances, the first test we used was Serious Sam First Encounter.

The highest scores here were recorded by the WinFast NF4K8MC. This nForce4-based